

Project Report

Self-Supervised Few-Shot Anomaly Detection in Time-Series Data

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Introduction:

This project implements a self-supervised learning (SSL) pipeline for few-shot anomaly detection in time-series data using an LSTM-Transformer backbone and a Siamese Network. The goal is to detect anomalies with minimal labeled data.

Dataset & Preprocessing:

Initially, the SMD dataset was selected and data loading pipelines were implemented for server machine training files. Then NAB dataset was used for full experimentation which consists of timestamped univariate time-series values. Data was loaded, timestamps converted, normalized, sequences of length 24 generated, and 15% of steps masked for self-supervised learning.

Methodology:

1. **Masked Time-Series Modeling:** Learns temporal patterns by predicting masked steps without labels.
2. **LSTM-Transformer Pretraining:** LSTM captures local, Transformer global dependencies, trained 20 epochs using MSE on masked positions.
3. **Few-Shot Siamese Learning:** Ten normal sequences form a support set; embeddings fine-tuned via centroid-distance loss for tight normal clustering.
4. **Anomaly Scoring:** Test embeddings' Euclidean distance from the centroid indicates anomalies; higher distance means anomaly.

Results:

Anomalous sequences scored higher than normal ones, validating the method.

Conclusion:

The SSL-based few-shot pipeline reliably detects anomalies using masked modeling, LSTM-Transformer embeddings, Siamese fine-tuning, and distance-based scoring, and can scale to larger datasets.